

BEE 208 — Question 14: Laboratory Login Attempt

Control Statement: while loop with a counter | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables accessCode (string), attempts (int), granted (bool)

Step 3: Set attempts = 0, granted = false

Step 4: While attempts < 3, prompt user to enter access code

Step 5: Read accessCode, increment attempts by 1

Step 6: If accessCode == 'BEE208', set granted = true, break

Step 7: Else display remaining attempts

Step 8: End loop

Step 9: If granted, display 'Access granted.'

Step 10: Else display 'Access denied. Maximum attempts reached.'

Step 11: End

2. PSEUDOCODE

```
BEGIN

DECLARE accessCode AS STRING

DECLARE attempts AS INTEGER

DECLARE granted AS BOOLEAN

SET attempts = 0, granted = FALSE

WHILE attempts < 3 DO

    OUTPUT 'Enter Laboratory Access Code:'

    INPUT accessCode

    attempts = attempts + 1

    IF accessCode == 'BEE208' THEN

        SET granted = TRUE

        BREAK

    ELSE

        OUTPUT remaining attempts

    END IF

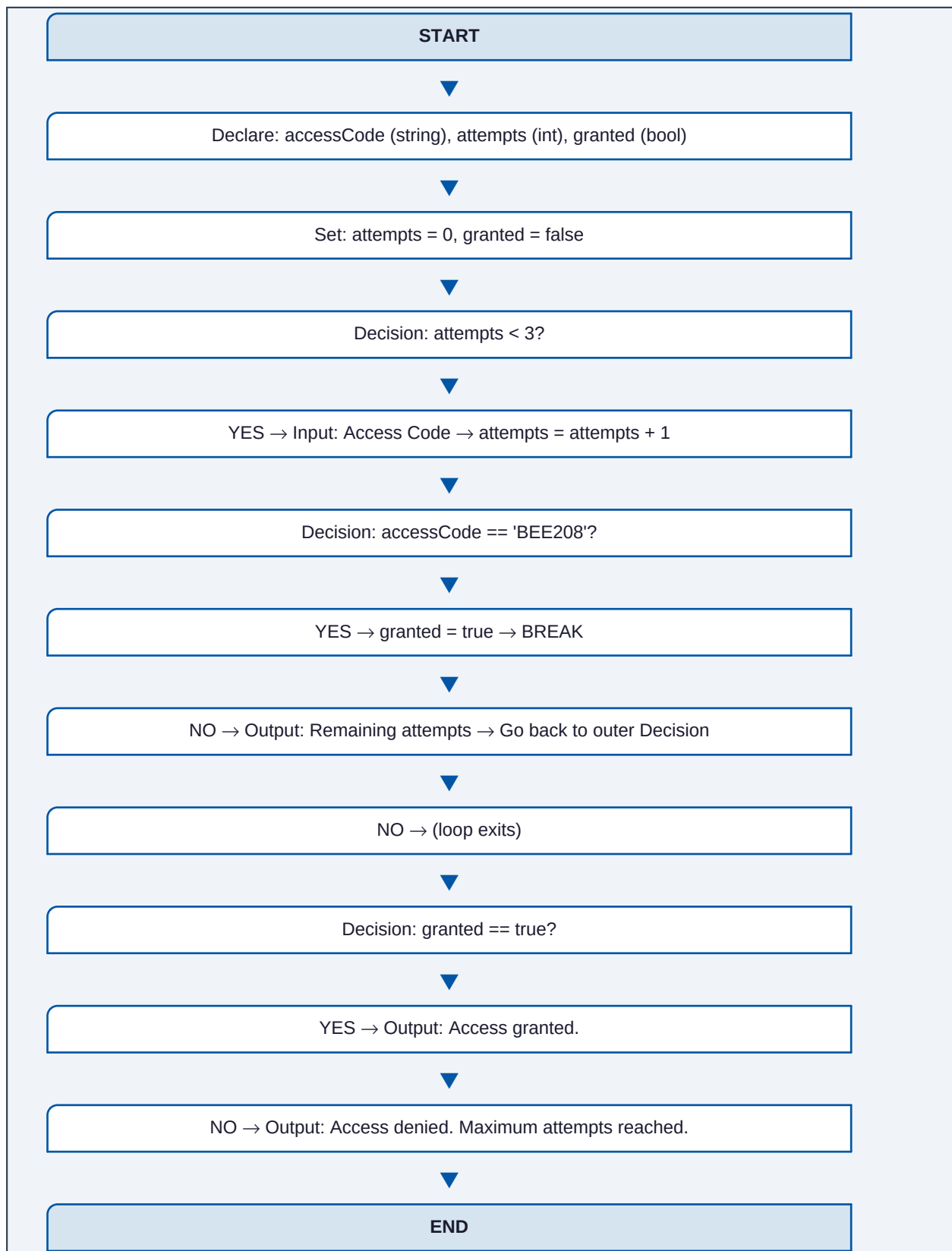
END WHILE

IF granted THEN OUTPUT 'Access granted.'
```

```
ELSE OUTPUT 'Access denied. Maximum attempts reached.'
```

```
END
```

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	Attempt 1: BEE208	Expected Output	Access granted.
Input	3 wrong codes	Expected Output	Access denied. Maximum attempts reached.